

Varun Sharma

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EDUCATION

Bennett University

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)

Greater Noida, UP

Expected May 2028

- CGPA: 7.4/10.0
- Relevant Coursework: Data Structures & Algorithms, AI, Machine Learning, Linear Algebra, OOP

TECHNICAL SKILLS

Programming Languages: Python (Proficient), C++ (DSA), TypeScript, JavaScript

Web & Backend: FastAPI, React, MongoDB, REST APIs

Data Science & AI/ML: NumPy, Pandas, Scikit-learn, Matplotlib, Google Gemini API

Developer Tools: Git, GitHub, GitHub Actions, Vercel, VS Code, Jupyter Notebooks

Core Competencies: Data Wrangling, Time-Series Analysis, Predictive Modeling, Backtesting, Risk Analytics

TECHNICAL PROJECTS

Algo Trade Simulator | *Python, FastAPI, React, TypeScript, MongoDB, Gemini API*

2026

[Live demo](#) · [Code](#) · Team project with Yashika Garg

- Built a full-stack backtesting platform that simulates trades on real market data with fees, slippage and no look-ahead bias, benchmarked against buy & hold and the S&P 500 on Sharpe, Sortino, drawdown and beta.
- Implemented walk-forward testing (tune on the past year, trade the next quarter on unseen data), which showed a tuned strategy's return falling from 24% to 10% a year out of sample.
- Developed a logistic-regression trading strategy from scratch with time-aware retraining, evaluated against a majority-class baseline and ROC-AUC.
- Integrated Gemini function calling so the AI copilot runs backtests and simulations; deployed on Vercel with Google OAuth and 107 automated tests in CI.

RazorGrowth: AI Merchant-Growth Agent | *Razorpay AI Buildathon 2026*

2026

- Built a permissioned AI agent that detects failed-payment recovery opportunities, identifying 5,495 in recoverable revenue from 7,850 in failed transactions using an LLM reasoning layer decoupled from financial execution authority.
- Designed a deterministic policy engine enforcing budget caps and action-type rules, blocking out-of-bounds AI proposals before human review or API execution.
- Implemented idempotent Razorpay REST integration with unique idempotency keys and retry-safe execution, guaranteeing zero duplicate transactions under network failure or timeout conditions.
- Shipped a production deployment (Vercel + Render) with 11 automated unit tests covering the policy engine, idempotency guarantees, and webhook signature verification (HMAC SHA256).

AgriGuard: Satellite Imagery Crop Analysis | *Python, Scikit-learn, Sentinel-2 Data*

2026

- Developed a classification system to identify crop types and detect diseases using multispectral imagery.
- Implemented image preprocessing and feature extraction on large-scale Kaggle datasets.

Sudoku Solver: Automated Logic Engine | *Python*

- Built an automated puzzle-solving application using a Recursive Backtracking algorithm.
- Optimized constraint logic for efficient solving of 9×9 grids.

LEADERSHIP & EXPERIENCE

Management Lead

Sportikon 3.0

2025

Peer Mentor

DSA Summit

2025

HONORS & ACTIVITIES

- **Open Source:** Maintaining public repos for AI/ML and algorithmic research.
- **Hackathons:** Registered participant for 2026 Inter-College events.